Screening for Type 2 Diabetes and Hypertension from Photoplethysmography with Wavelet Images and Vision Transformers

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Code: `https://github.com/Suraba03/bsc-thesis`

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Abstract

Type 2 diabetes and arterial hypertension usually remain asymptomatic until complications appear, while the standard diagnostic tests require a clinic visit. Photoplethysmography (PPG) is a cheap optical signal that wearables already record continuously, and both diseases are known to change the shape of the PPG pulse wave. We study whether a single pipeline can estimate type 2 diabetes risk and blood pressure from raw PPG. Signals from the MIMIC-III clinical database are band-pass filtered, converted into time–frequency images with the continuous wavelet transform (CWT), and passed to an image model. We compare a modified VGG network with a Swin Transformer, both initialised from pre-trained weights. On a hold-out set, the Swin Transformer reaches an AUC-ROC of 0.72 and an AUC-PRC of 0.68 for diabetes classification (VGG: 0.62 and 0.55) and a mean absolute error of 8.5 mmHg for systolic blood pressure (VGG: 10.5 mmHg); VGG is slightly better for diastolic pressure (5 vs. 6 mmHg). These results are not accurate enough for diagnosis, but they support the use of transformer models on CWT representations of PPG as a preliminary screening tool. We also release a web prototype that runs the full pipeline on an uploaded recording.

Keywords: photoplethysmography, diabetes, hypertension, continuous wavelet transform, Swin Transformer, wearable health

1 Introduction

Type 2 diabetes and arterial hypertension are among the most common chronic diseases. Both often go unnoticed until complications develop: retinopathy, nephropathy, neuropathy and cardiovascular disease in the case of diabetes, and stroke, myocardial infarction, heart failure and chronic kidney disease in the case of hypertension. The established tests, the HbA1c blood test for diabetes and repeated or ambulatory blood-pressure measurement for hypertension, cost time and money, so many people are not screened regularly.

Photoplethysmography (PPG) measures changes in blood volume in small vessels over the cardiac cycle with a light source and a photodetector [1, 2]. It is non-invasive, inexpensive and already built into fitness bands and smartwatches. Both diseases leave traces in the PPG waveform: hypertension through increased arterial stiffness, which lengthens the rising edge and weakens the diastolic peak, and diabetes through changes in microcirculation and endothelial function. A screening tool based on PPG could therefore run continuously on devices that many people already wear. At the time of this work, however, deep-learning studies of PPG-based diabetes detection were limited to roughly a dozen papers.

This work addresses four questions:

1. **Diabetes risk.** Can the risk of type 2 diabetes be predicted from a raw PPG recording by filtering the signal, turning it into a CWT image and classifying the image with a deep network?

2. **Blood pressure.** Can the same pipeline estimate systolic and diastolic blood pressure, so that hypertension can be inferred from standard pressure categories?

3. **Architecture.** Does a hierarchical vision transformer (Swin Transformer), which models both local patterns and long-range dependencies, outperform a convolutional network on CWT images of PPG?

4. **Deployment.** Can the models be packaged into an application that accepts a user's recording, shows each processing step and returns risk estimates in real time?

2 Related work

From hand-crafted features to deep learning. Early work on non-invasive disease detection from optical signals relied on features designed from domain knowledge. Yamakoshi et al. [3] estimated blood glucose with multivariate regression and classification models on "pulse glucometry" data. Deep learning removed much of this manual feature engineering: Shimazaki et al. [4] learned features for cuffless blood-pressure estimation from the PPG waveform with an autoencoder, and Srinivasan and Foroozan [5] built a deep-learning diabetes predictor from PPG that uses the continuous wavelet transform to obtain more informative inputs, the approach most directly followed here. Nakamura and Furukawa [6] also used CWT processing to characterise glucose changes in diabetes, and Jiang et al. [7] identified diabetes risk factors with interpretable machine learning on census data. For hypertension and cardiovascular disease, Nuryani et al. [8] detected hypertension without a cuff using a swarm-optimised support vector machine on PPG and ECG, and Sadek et al. [9] detected cardiovascular disease from PPG with machine learning. Charlton et al. [2] review wearable PPG for cardiovascular monitoring.

Learning with little labelled data. Labelled physiological data are scarce, and self-supervised learning [10] is increasingly used to reduce the need for labels. Akamatsu et al. [11] measure heart and respiratory rate from video without explicit labels by calibrating between multiple cameras, and Hashimoto et al. [12] use self-supervised feature extraction from resting-state fMRI for schizophrenia diagnosis.

Difficulties specific to PPG. Three obstacles recur in the literature. First, PPG is a much coarser view of cardiac activity than ECG. Second, the waveform is also shaped by other conditions, such as vascular ageing and atherosclerotic plaques. Third, reliable labels are hard to obtain because a patient may have several conditions at once: a past myocardial infarction changes the PPG of a patient who may still be labelled "no diabetes". Proposed remedies include combining PPG with other biomarkers, self-supervised pre-training, individualised models and better sensors.

3 Data

We set the dataset requirements before the search: at least 1,000 subjects, several recordings per subject, basic demographics (sex and age), a medical history that is as complete as possible, and a sampling rate of at least 125 Hz. The MIMIC-III clinical database [13] meets these requirements and was used for all experiments: it contains repeated PPG recordings from more than 1,000 subjects together with sex, age and medical history. Signals sampled at 125 Hz were preferred over higher sampling rates, which were noisier in our experiments. Waveforms were read with the WFDB software package [14–16], and physiological signal processing relied on NeuroKit2 [17, 18]. The medical history and blood-pressure records in the database provide the diabetes labels and the regression targets.

4 Method

Filtering. We implemented Butterworth and Chebyshev band-pass filters and compared them experimentally. A third-order Butterworth filter with a 0.5–8 Hz pass band removed both high-frequency noise and low-frequency baseline wander, such as the respiratory component, and was used in the final pipeline. The fast Fourier transform and its inverse were used to move between the time and frequency domains. Healthy PPG signals were also generated synthetically to augment the training data.

Time–frequency images. Each filtered segment $x(t)$ is converted into a two-dimensional image with the continuous wavelet transform

$$W_x(a,b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} x(t)\, \psi^*\left(\frac{t-b}{a}\right) dt, \tag{1}$$

where $a > 0$ is the scale, b the time shift and ψ a Gaussian wavelet that satisfies the finite-energy and admissibility conditions. The magnitude $|W_x(a,b)|$ over a grid of scales and shifts forms the input image. Augmentations were applied to the CWT images to reduce overfitting.

Models. Two image models were trained for both tasks, diabetes classification and blood-pressure regression: a modified VGG network and a Swin Transformer, each initialised from weights pre-trained on natural images. Both used a batch size of 32 and a learning rate of $3 \cdot 10^{-4}$. The VGG model was trained with SGD with momentum and cosine annealing, and the Swin Transformer with AdamW and learning-rate warm-up. L2 regularisation and weight

decay were used against overfitting. Each model was trained for 20 epochs; the regression models were trained with a mean squared error (MSE) loss.

Evaluation. The system is meant as a screening aid, so for classification we report AUC-ROC and AUC-PRC, which reflect the trade-off between recall and false positives. For blood-pressure regression we report the mean absolute error (MAE) in mmHg. All metrics are computed on a hold-out set.

Application. A Flask web application loads the trained weights, accepts a PPG recording, shows each stage of the pipeline (raw signal, filtered signal, CWT image) and returns the diabetes risk, the predicted systolic and diastolic pressure and the corresponding hypertension category.

5 Results

Table 1 reports diabetes classification and Table 2 blood-pressure estimation.

Table 1: Type 2 diabetes classification on the hold-out set: best values over 20 training epochs and the epoch with the lowest loss.

Model	Loss	AUC-ROC	AUC-PRC	Epoch
VGG (modified)	0.5	0.62	0.55	10
Swin Transformer	**0.3**	**0.72**	**0.68**	18

Table 2: Blood-pressure estimation on the hold-out set: best MSE loss and MAE (mmHg) over 20 epochs, and the epoch with the lowest loss. Lower is better.

	Systolic			Diastolic		
Model	MSE	MAE	Epoch	MSE	MAE	Epoch
VGG (modified)	50	10.5	13	35	**5**	13
Swin Transformer	**45**	**8.5**	15	**30**	6	10

The Swin Transformer trained more stably and was better for diabetes classification and systolic pressure, while VGG was slightly better for diastolic pressure. This supports the architecture hypothesis for most targets, and the Swin Transformer was therefore used in the web application.

6 Discussion and limitations

The metrics are not good enough for diagnosis. An AUC-ROC of 0.72 is useful only as a signal for referring someone to a proper test, and a systolic MAE of 8.5 mmHg is above the error usually tolerated for blood-pressure devices. We see three main reasons. The labels come from clinical records of intensive-care patients, who often have several conditions that also affect the PPG waveform, so the labels are noisy with respect to the PPG signal. MIMIC-III patients are not representative of the healthy population that a wearable screening tool would target. The reported numbers are the best values reached on the hold-out set during training, without a separate validation set for model selection, so they are optimistic estimates. Finally, the synthetic healthy signals used for augmentation may differ from real recordings in ways the models can exploit. A per-subject split, external validation on wearable data and calibration analysis would be needed before any practical use.

7 Conclusion and future work

We built a complete pipeline from raw PPG to diabetes-risk and blood-pressure estimates: band-pass filtering, CWT images, convolutional and transformer models compared on hold-out data, and a working web prototype. Natural next steps are larger and better-labelled datasets, combining PPG with other biomarkers, self-supervised pre-training to reduce the dependence on scarce labels, and individualised models.

Code and data availability

Code, trained-model loading and the web application are available at `https://github.com/Suraba03/bsc-thesis`. MIMIC-III is distributed by PhysioNet under a credentialed-access licence and is not redistributed with the code.

Note

This preprint is a condensed English version of the author's bachelor's thesis (Moscow Aviation Institute, 2024; written in Russian). It has not been peer reviewed.

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